Survival Analysis of Pediatric Bone Marrow Transplant Patients

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1 Introduction

This lab analyses the survival of pediatric patients who underwent bone marrow transplantation. The dataset contains 187 patients and 37 variables, including demographic, clinical and transplantation-related features. The main outcome variables are survival time and survival status, where a status of one indicates that the event occurred and zero indicates a censored observation.

2 Exploratory Analysis

The dataset contains 85 observed events and 102 censored observations. Missing values were found mainly in categorical transplantation-related variables and in some recovery measurements. In addition, the value 1000000 was interpreted as an indicator that a recovery or complication had not been observed and was therefore treated as a missing value during preprocessing.

The numerical variables with the strongest correlation with survival time were recipient body mass ($r = -0.145$) and recipient age ($r = -0.138$). These correlations were weak, suggesting that survival time cannot be explained by a single numerical variable. The strongest correlations between input features were observed between recipient age and body mass ($r = 0.898$), and between CD34 cell dose and CD3 cell dose ($r = 0.583$). Due to this redundancy, the features were standardized before dimensionality reduction.

Categorical variables were one-hot encoded and missing values were imputed using the median of the corresponding feature. This resulted in 83 input features without missing values.

3 Clustering

Principal Component Analysis was applied after standardization. A total of 29 principal components were required to preserve approximately 90.1% of the original variance.

K-means clustering was evaluated for different numbers of clusters using the silhouette score. Three clusters were selected because this configuration provided the highest silhouette score among the tested alternatives. The resulting cluster sizes were 28, 94 and 65 patients.

The clusters showed only moderate differences in their average survival times. Their mean survival times were approximately 1098, 903 and 922 days, respectively. This suggests that the clustering captured differences in the patients' input characteristics, but these differences did not necessarily translate into clearly different survival outcomes.

4 Survival Analysis

Kaplan–Meier survival curves were estimated separately for each cluster. The curves overlapped substantially, and the median survival time was not reached in any cluster because the estimated survival probability did not fall below 0.5 during the observation period. A multivariate log-rank test showed no statistically significant difference between the clusters ($p = 0.89$).

A Cox proportional hazards model using the cluster labels as predictors produced hazard ratios of 0.87 and 0.92 for clusters 1 and 2 relative to cluster 0. However, both effects were non-significant ($p = 0.65$ and $p = 0.81$). The concordance index was 0.52, indicating that the cluster labels alone had almost no predictive ability for ranking patients according to their survival risk.

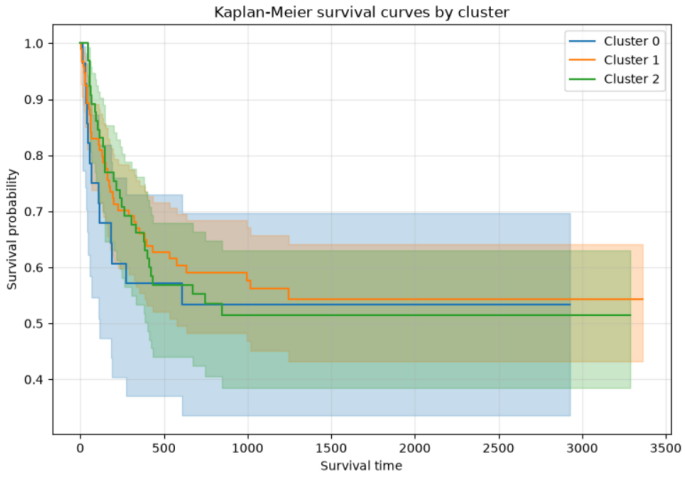


Figure 1: Kaplan–Meier survival curves for the three patient clusters.

A second Cox model was fitted using clinical and transplantation-related variables. The proportional hazards assumption was violated by the relapse variable, so relapse was included as a stratification variable in the final model. This stratified model obtained a concordance index of 0.69, which represents moderate discrimination.

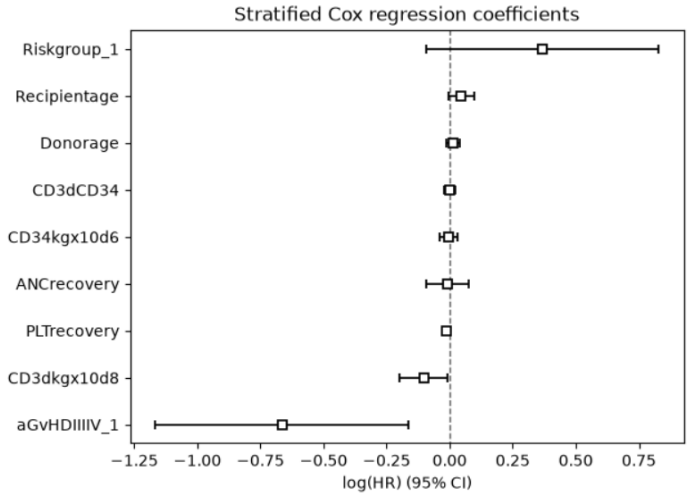


Figure 2: Coefficients and 95% confidence intervals for the stratified Cox model.

The most relevant variables were CD3 cell dose, platelet recovery and acute graft-versus-host disease. CD3 cell dose and platelet recovery were associated with lower hazard ratios, while the presence of acute graft-versus-host disease showed a hazard ratio of approximately 0.51. Relapse was also highly relevant in the initial model, although it was not assigned a coefficient in the stratified model because it was used to define separate baseline hazards.

5 Conclusions

The unsupervised clusters did not produce statistically significant differences in survival. Therefore, clustering patients according to the available input features was not sufficient to identify groups with clearly different survival outcomes.

The Cox model based on clinically meaningful variables performed better than the cluster-based model, with a concordance index of 0.69 compared with 0.52. This suggests that individual clinical variables contain more useful prognostic information than the unsupervised clusters. Nevertheless, it also important to consider that the dataset is samll and therefore the results have to be interpreted cautiously.

The code used for the analysis is available at:

`https://github.com/Jacobotaboadaesteban/lab2-survival-analysis`